

MLB Catcher Framing Analytics Platform

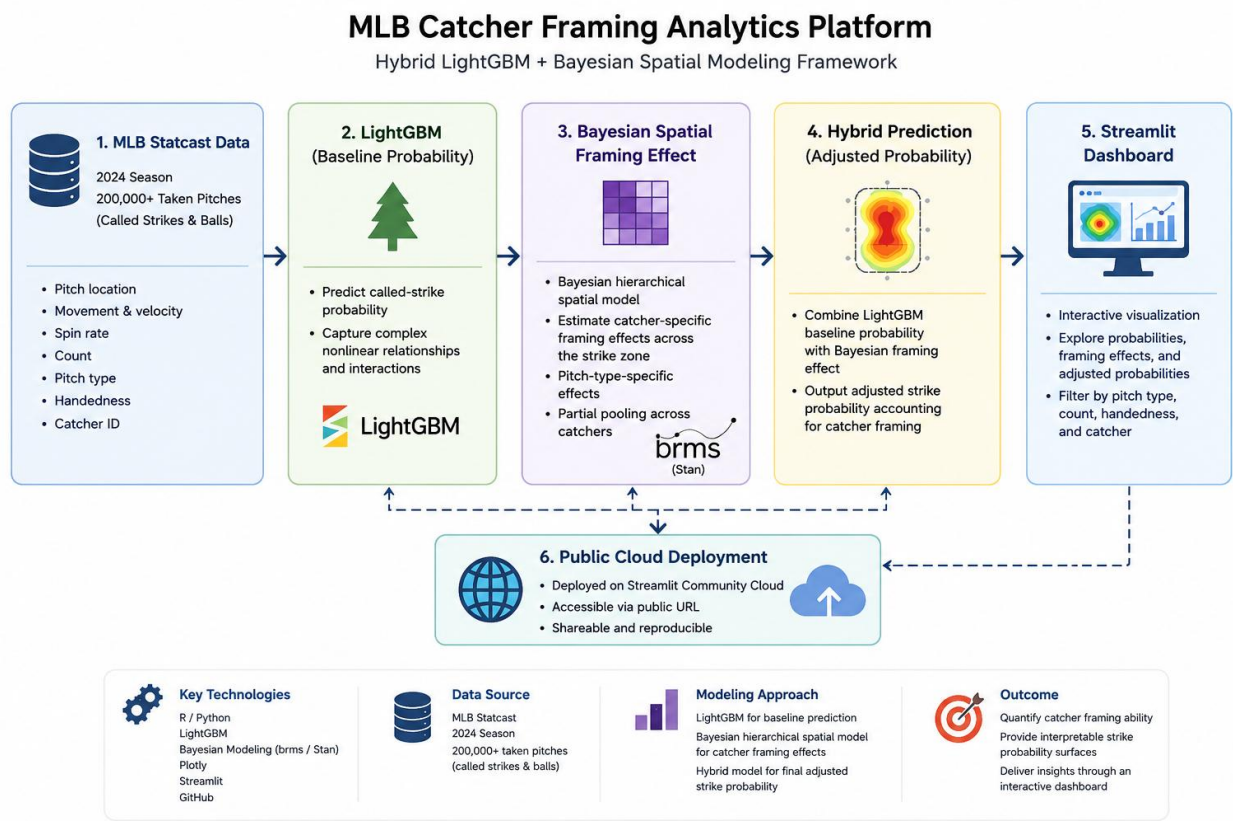
A Hybrid LightGBM and Bayesian Spatial Modeling Framework

1. Project Overview

This project develops an interactive analytics platform for evaluating MLB catcher framing effects. The framework combines machine learning and Bayesian hierarchical modeling to estimate called-strike probability and quantify catcher-specific framing effects across the strike zone.

Objectives

- Estimate baseline called-strike probability.
- Quantify catcher-specific framing effects.
- Build an interpretable hybrid prediction framework.
- Deploy an interactive web dashboard for exploration.



2. Dataset and Modeling Framework

Dataset

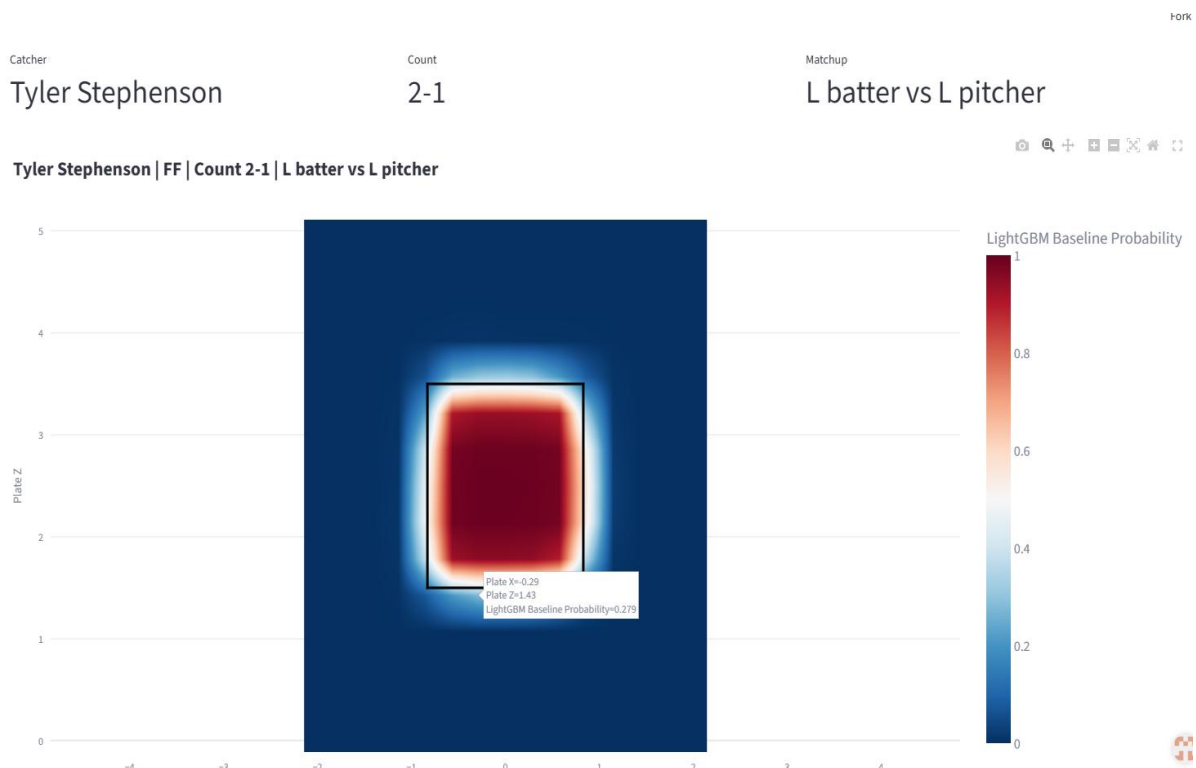
Source: 2024 MLB Statcast pitch-level data

Scope: Taken pitches only (called strikes and called balls)

Size: Over 200,000 pitch observations

LightGBM Baseline Model

- Pitch location (plate_x, plate_z)
- Pitch movement (pfx_x, pfx_z)
- Pitch velocity
- Spin rate
- Ball-strike count
- Pitch type
- Batter handedness
- Pitcher handedness
- Catcher identity

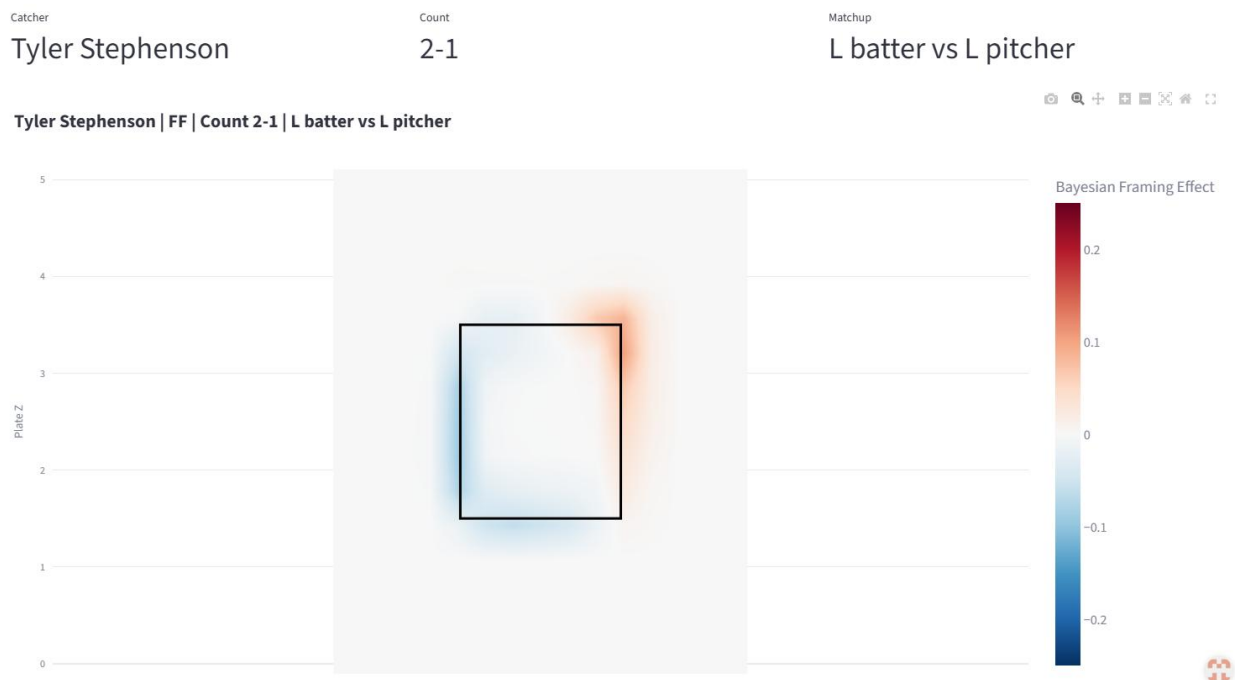


3. Bayesian Spatial Framing Model

A Bayesian hierarchical spatial model was developed to estimate catcher-specific framing effects across the strike zone.

- Spatial strike-zone surfaces
- Pitch-type-specific spatial effects
- Catcher-specific framing surfaces
- Hierarchical partial pooling across catchers

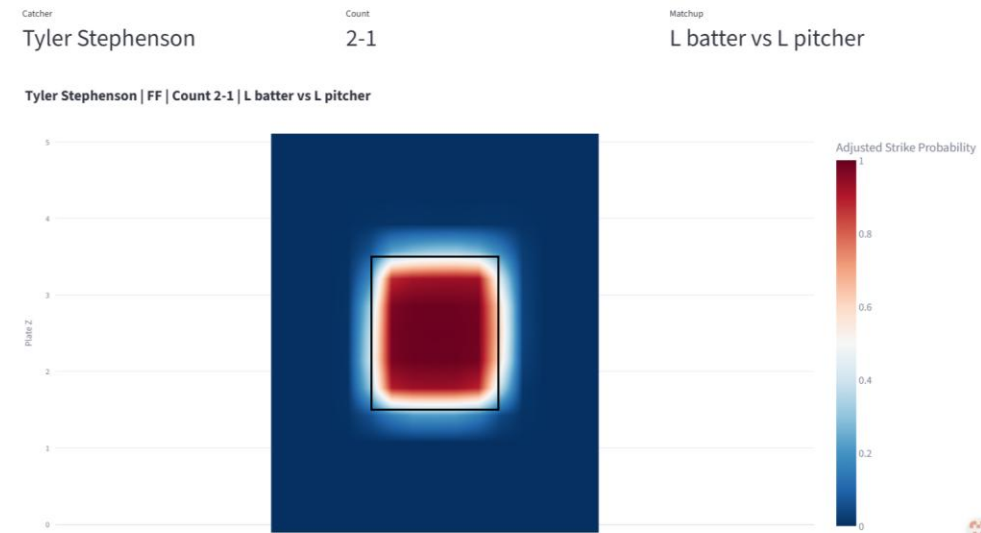
Posterior estimates quantify framing effects beyond the baseline strike probability predicted by LightGBM.



4. Hybrid Prediction and Dashboard Deployment

The final prediction combines the LightGBM baseline probability with Bayesian framing effects to generate an adjusted strike probability surface.

Adjusted Probability = LightGBM Baseline Probability + Bayesian Framing Effect



Dashboard Features

- Interactive catcher selection
- Pitch-type filtering
- Count-based analysis
- Batter/pitcher handedness matchup analysis
- Dynamic heatmap visualization

Technical Stack

- R
- Python
- LightGBM
- Bayesian Hierarchical Modeling
- brms / Stan
- Plotly
- Streamlit
- GitHub

Links

Live Demo: <https://mlb-bayesian-framing-dashboard-xrklc7ou9vak2boyibczh3.streamlit.app/>

GitHub Repository: <https://github.com/Samuel0417/mlb-bayesian-framing-dashboard>